
Six math clues · answer key

Shadows at Noon

Age 8 / Grade 2 stretch. Offer tens-and-ones drawings, repeated addition, or oral reasoning.

Math 1 · 8

2, 4, 6, 8. Add 2 to 6 to reach 8.

Math 2 · 20

5, 10, 15, 20. Add 5 to 15 to reach 20.

Math 3 · 40

10, 20, 30, 40. Add 10 to 30 to reach 40.

Math 4 · 14

12, 14, 16, 18. The missing number is 14.

Math 5 · 35

25, 30, 35, 40. The missing number is 35.

Math 6 · 40

40, 50, 60, 70. The missing number is 40.

Age 8 / Grade 2 math. Six skip-counting sequences use steps of 2, 5, and 10. These number trails practice counting; they do not claim that shadows change at a steady rate. Welcome spoken reasoning or marks on paper.

Reading & writing · answer key

Shadows at Noon

Reading · 1

The sun was higher around noon, so the same standing stick cast a shorter shadow.

Reading · 2

The shadow points away from the light. The bright side of the stick faces the sun.

Reading · 3

Leave the stick standing in the same place. Mark each shadow tip, measure from the stick with the same unit, and compare the lengths.

Writing

Accept one sensible command, spoken, dictated, or written. Example: "First stand the stick, then mark the tip of the shadow."

Age 8, Grade 3 reading, and Grade 2 math. This story uses a sunny yard at middle latitudes. Solar noon is when the sun reaches its highest point that day; it need not be 12:00 by the clock. For the same vertical stick on level ground, that gives the shortest daytime shadow. Height varies with place and season; noon is not always overhead. Clear sunlight gives a distinct edge; clouds can soften it. The diorama isolates sun height and light direction, with equal chalk spaces for comparison, not a clock or local forecast. The number trails do not model a steady rate of shadow change. For an optional outdoor try, use a stable blunt stick beside a path with an adult, and study the ground rather than looking at the sun.

Accept a spoken command, dictation, invented spelling, or writing. One clear sentence is enough. Example: "First stand the stick, then mark the tip of the shadow."

Four shadow clues

Shadows at Noon

Quiz 1. In this sunny yard, what happens to the same stick's shadow around noon?

Around noon, the sun is near its highest point for the day, so the same standing stick has a shorter shadow.

Quiz 2. What happens to the shadow when the sun is lower?

A lower sun makes a longer shadow from the same standing stick.

Quiz 3. Which way does a shadow point?

A shadow points away from the light. The bright side of the stick faces the sun.

Quiz 4. How can Little compare morning and noon shadows?

Keep the stick standing in the same place. Mark each tip and measure from the stick with the same unit, then compare the lengths.

Age 8, Grade 3 reading, and Grade 2 math. This story uses a sunny yard at middle latitudes. Solar noon is when the sun reaches its highest point that day; it need not be 12:00 by the clock. For the same vertical stick on level ground, that gives the shortest daytime shadow. Height varies with place and season; noon is not always overhead. Clear sunlight gives a distinct edge; clouds can soften it. The diorama isolates sun height and light direction, with equal chalk spaces for comparison, not a clock or local forecast. The number trails do not model a steady rate of shadow change. For an optional outdoor try, use a stable blunt stick beside a path with an adult, and study the ground rather than looking at the sun.